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BCS-6T

Q1) Fitt's law predicts the time T to acquire a target as $MT = a + b\log_2\left(\frac{2A}{W}\right)$

where MT : to move & select (target)
W : width
A : Amplitude/Distance
a, b } constants

The concept of infinitely large target refers to target located at the edges & corners of the screen, such as start button in windows, the effective width in that direction of these targets are infinite no matter how much further user moves the mouse making them fastest target to acquire preventing overshooting of cursor.

Since fitt's law was one dimensional originally, researchers proposed models to interpret width for 2D targets.

1. STATUS QUO - horizontal extent
2. W+H - Sum of width & height
3. WXH - Area
4. SMALLER OF - smaller of width & height
5. W' - Distance across the target along line of cursor movement.

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Q2) Inductive reasoning is a bottom-up logical process where specific observations, experience & pattern to form a broader generalization.
for example $2+4=6$, $8+2=10$
Hypothesis: Sum of any two even integers is always even

Deductive reasoning is top-Down approach that moves from general premises to specific, guaranteed solutions.

Proof: $\left.\begin{array}{l} A: 2n \\ B: 2m \end{array}\right\}$ given 2 even integers $\longrightarrow$ $\underbrace{2(n+m)}_{\text{sum will always be even}} \rightarrow 2\overset{n+m}{K}$

Q3) → Total length = 14

Chunk length	No. of chunks	Memo time	Cognitive load
2	7	130s	High
3	5	109s	Moderate
4	4	72s	light
5	3	61s	light
7	2	140	High
14	1	180	High

Best: 5 chunk length, it was comprehendable with the brain capacity & cognitive load was less, it also follows miller's law range. (7±2).

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Q4) Monthly interface updates disrupts the process of skill acquisation users are prevented from reaching the autonomous stage of learning where little cognitive effort is required. The users are constantly forced back to cognitive stage, requiring conscious effort to navigate the new layout & prevent the layout map in LTM.

* HCI Problems:
  1. Inconsistency
  2. High cognitive load
  3. Prevention in shifting to LTM
* Better Design Approach
  1. Keep core navigation elements intact
  2. Add tutorials/user guide in startup
  3. Implement bit by bit change over time
  4. User controlled layout (Favorites/wishlist)

Q5) a)
- Long multipage form overwhelms user processing capacity due to cognitive overload
- Less attention span cause errors in lengthy tasks as focus reduces with time
- User cannot see all at once, making it difficult to plan ahead leading to more back again causing frustration
- Limited STM cause data prone entry as they cant remember what they entered in page 1

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b) • Auto save on cloud or device to let user fill it in sessions
• Progress indicator to show number of fields left to give sense of advancement
• Auto fill most of the fields by scanning documents to prevent manual entry & fast form fillups
• Group related sections together to prevent wrong data inputs
• Reduce manadatory fields as much as possible
• Realtime validate the fields for formats for example @ for email, dashes for CNIC
• Reduce text field input as much as possible to reduce typing time & errors. Add checkboxes, drop down, radiobuttons where possible

CamScanner